SUEZ Platform

AWS-Based High-Level Implementation Stack

A reference implementation for a governed enterprise AI gateway and control plane

Document intent
Translate the SUEZ product requirements into a practical AWS implementation stack that is detailed enough for architecture review, but high level enough to guide phased delivery.

Target outcome

• Provide secure multi-tenant AI access across providers such as OpenAI, Anthropic, Amazon Bedrock, Vertex AI, and Azure OpenAI.

• Separate the **control plane** for provisioning and policy management from the **data plane** that processes live AI requests.

• Meet enterprise requirements for tenant isolation, auditability, cost visibility, observability, and resilient SaaS operations.

• Create a stack that can start small for ~100 tenants and evolve toward ~10,000 tenants and ~1,000 requests/second.

Reference architecture posture

Layer	Implementation detail
Tenant & admin experience	Admin portal on S3 + CloudFront or containerized UI on EKS; enterprise SSO; RBAC-backed admin workflows.
Control plane	Tenant management, provider onboarding, model / tool / agent provisioning, policy authoring, cost profile setup, billing setup.
Data plane	AI gateway on EKS for request admission, policy evaluation, provider selection, token quota checks, prompt filters, audit emission, and response shaping.
Execution plane	Agent runtime on EKS, Step Functions for long-running orchestration, SQS buffers for retries, Lambda for lightweight task handlers.
Data & analytics plane	Aurora for transactional metadata; DynamoDB for high-throughput state; S3 data lake for usage events; Athena / QuickSight for finance and ops views.
Security & operations	WAF, GuardDuty, Security Hub, CloudTrail, KMS, Secrets Manager, CloudWatch, incident alerting, and immutable audit retention.

Design assumption: latency-sensitive gateway functions run on Amazon EKS, while highly asynchronous work such as metering aggregation, billing enrichment, and long-running agent flows moves to queue- or workflow-driven services.

1. Core AWS Service Mapping

Capability	Recommended AWS Services	Purpose
Edge / ingress	Amazon Route 53, AWS WAF, AWS Shield, Amazon CloudFront, ALB	Global DNS, DDoS protection, L7 filtering, TLS termination, request routing
Identity	Amazon Cognito or federation to enterprise IdP, AWS IAM, IAM Identity Center, Secrets Manager	User / admin auth, service identities, temporary credentials, secret rotation
API & control plane	Amazon EKS, AWS App Mesh (optional), API Gateway (admin APIs)	Host stateless microservices for tenant, policy, model, tool, agent, and admin APIs
Data plane / gateway	Amazon EKS, ALB, Amazon ElastiCache for Redis	Low-latency policy checks, provider routing, quotas, cache, session and request coordination
Agents & async jobs	Amazon EKS, Amazon SQS, AWS Step Functions, AWS Lambda	Background workflows, retries, queue-based agent steps, event-driven jobs
Storage	Amazon Aurora PostgreSQL, DynamoDB, S3, EFS (optional)	Transactional metadata, key-value state, audit / prompts / exports, shared artifacts
Events & metering	Amazon MSK or EventBridge + Kinesis, AWS Glue, Athena	Usage events, billing pipeline, stream processing, analytics and reporting
Observability	Amazon CloudWatch, AWS X-Ray / OpenTelemetry, CloudTrail, Security Hub, GuardDuty	Metrics, logs, traces, audit, threat detection and posture visibility
CI/CD & IaC	CodePipeline / GitHub Actions, ECR, AWS CDK / Terraform	Build, scan, publish containers, promote environments, manage infrastructure

2. High-Level Request Flow

Nominal transaction path

1) Enterprise application authenticates the caller via SSO / enterprise identity and sends the request to the SUEZ gateway endpoint behind Route 53, CloudFront, WAF, and ALB.
2) Gateway services on EKS resolve tenant context, load policy and quota state from Redis / Aurora, and validate model, tool, provider, and prompt-size rules.
3) If the request targets an agent, the agent runtime coordinates model calls and approved tool invocations; long-running or parallelizable steps are pushed to SQS or Step Functions.
4) The gateway calls the selected AI provider using provider-specific adapters, with secrets retrieved from Secrets Manager and encrypted by KMS.
5) Usage events are emitted to Kinesis / MSK / EventBridge, persisted to S3 and analytics stores, then transformed for billing, operations, and compliance reporting.
6) Metrics, structured logs, traces, and security events are shipped into CloudWatch, OpenTelemetry collectors, CloudTrail, GuardDuty, and Security Hub.

3. VPC and Network Topology

Network segmentation recommendations

Use a multi-AZ VPC with isolated **public ingress**, **private application**, and **data** subnets. Keep EKS nodes, Aurora, Redis, MSK, and internal services in private subnets only.
Expose only ALB / CloudFront to the internet. All east-west calls remain private and security-group restricted.
Use VPC endpoints / PrivateLink where supported to reduce internet egress for AWS-native dependencies.
Adopt separate AWS accounts for shared services, non-production, and production. For larger scale, split by environment and optionally by region using AWS Organizations and SCP guardrails.

4. Security, Identity, and Tenant Isolation

• **Identity:** Federate administrators and enterprise users from the customer IdP; use IAM roles for workloads; prefer short-lived credentials over static keys.

• **Secrets:** Store provider credentials, tenant-scoped tool secrets, webhook tokens, and encryption material in Secrets Manager with automatic rotation where possible.

• **Encryption:** Use KMS for envelope encryption on Aurora, DynamoDB, S3, EBS, and Secrets Manager; consider separate keys per environment and, if needed, per regulated tenant class.

• **Tenant isolation:** Combine logical isolation in the data model with row-level or schema-level isolation in Aurora, tenant-partition keys in DynamoDB, and strong authorization in the gateway. Reserve physically isolated stacks for premium or regulated tenants.

• **Guardrails:** WAF managed rules, prompt / payload filters, outbound egress controls for tools, immutable audit logs, CloudTrail organization trails, and continuous findings via Security Hub + GuardDuty.

5. Data, Metering, and Billing Design

• Use **Aurora PostgreSQL** for tenant, user, model, tool, policy, billing-plan, and agent metadata that benefits from relational integrity and transactional updates.

• Use **DynamoDB** for high-cardinality request counters, idempotency keys, token quota snapshots, session state, and rate-limit ledgers.

• Use **S3** as the system of record for immutable audit exports, usage event archives, invoice inputs, operational snapshots, and analytics-friendly parquet datasets.

• Feed streaming usage into **Kinesis or MSK**; transform with Lambda, ECS jobs, or Glue; expose cost and usage dashboards through Athena and QuickSight.

6. Reliability and Operations

Operational area	AWS implementation guidance
Availability target	Run ALB, EKS worker nodes, Aurora, Redis, and streaming services across at least three Availability Zones.
Disaster recovery	Back up Aurora continuously, enable point-in-time restore, version S3 buckets, and replicate critical datasets to a secondary region if RPO / RTO requires it.

Operational area	AWS implementation guidance
Scaling	Use horizontal pod autoscaling on request-rate and latency metrics; decouple bursty work with SQS; right-size node groups or use Karpenter for efficient scaling.
Release strategy	Adopt blue/green or canary deployments for gateway services; scan images before release; gate production promotion with smoke tests and rollback hooks.
Observability	Define golden signals per service, trace provider adapters, create billing lag and policy-failure alarms, and preserve audit trails for compliance review.

Recommended first release: Route 53 + CloudFront + WAF + ALB; Amazon EKS for gateway and control-plane services; Aurora PostgreSQL; DynamoDB for quotas and idempotency; ElastiCache Redis for low-latency cache; S3 for audit and usage archives; EventBridge or Kinesis for metering; CloudWatch + CloudTrail + Security Hub for operations and governance.